Article

CLOP-DiT: Text-Guided Single-Cell Gene Expression Generation via Contrastive Language-Omics Pretraining and Diffusion Transformers

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Abstract

Generating realistic synthetic single-cell gene expression profiles conditioned on natural-language descriptions of cell identity would enable data augmentation and rare cell-type simulation, and could in future support hypothesis-driven perturbation studies. We present CLOP-DiT, a two-stage pipeline combining Contrastive Language–Omics Pretraining (CLOP) with a conditional Diffusion Transformer (DiT) trained via flow matching. CLOP aligns BiomedBERT text embeddings and scGPT cell embeddings into a shared 512-dimensional space (3.02M parameters), amplifying inter-type separation from 0.6% to 78% in cosine distance. DiT1D (22.10M parameters, 8 AdaLN-Zero blocks) then performs conditional flow matching in scGPT latent space, and a frozen scGPT decoder maps generated latents to gene expression. Across 100 cell types from 80 datasets (220,304 cells, 1,088 text groups), CLOP-DiT achieves k-nearest-neighbor accuracy of 36.9% (37$\times$ random chance, but 52 percentage points below the 89% real-data baseline) and steering accuracy of 81.0% (random = 50%) in the high-fidelity regime (CFG = 2.0, Euler solver); the recommended high-diversity regime (CFG = 1.0, Midpoint solver) reaches a diversity ratio of 0.93 at 80.7% steering. Unconditional generation collapses to random baselines, confirming that text conditioning is the sole driver of type specificity. In-distribution downstream biological validation—clustering alignment, classifier transfer (expression-space logistic regression accuracy 30.8%, macro F1 0.247), and differential expression concordance (mean logFC Pearson $r \approx 0.39$ across three contrasts, range 0.17–0.56)—demonstrates that generated cells partially integrate into standard single-cell workflows, though a discriminator AUC of 0.656 and the 52-point KNN gap from real data indicate substantial room for improvement.

Keywords: single-cell RNA-seq; generative model; diffusion transformer; contrastive learning; text-guided generation; flow matching; gene expression; foundation model

1. Introduction

Single-cell RNA sequencing (scRNA-seq) has enabled the characterization of cellular heterogeneity at unprecedented resolution [1]. However, generating realistic synthetic single-cell expression profiles conditioned on natural-language descriptions of cell identity remains an open challenge. Such a capability would enable data augmentation for rare cell types, and could potentially support hypothesis testing via *in silico* perturbation and atlas completion without additional sequencing, though these applications remain to be validated.

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Existing approaches to single-cell data generation include variational autoencoders such as scVI [2] and scGen [3], geometric deep generative models [4], and more recently, large-scale single-cell foundation models such as Geneformer [11] that leverage transfer learning for downstream prediction. These methods condition on categorical cell-type labels or perturbation metadata, but to our knowledge none accept structured natural-language biological descriptions as conditioning input. In parallel, contrastive vision–language models such as CLIP [5] have demonstrated that aligning two modalities into a shared embedding space enables powerful zero-shot transfer, and domain-specific variants such as BiomedCLIP [6] extend this paradigm to biomedical data. Meanwhile, Diffusion Transformers (DiTs) [7] have emerged as state-of-the-art conditional generators, and flow matching [8] provides a simulation-free training objective that simplifies the diffusion framework.

We present CLOP-DiT, a two-stage pipeline that combines **C**ontrastive **L**anguage–**O**mics **Pr**etraining (CLOP) with a conditional **Di**ffusion **T**ransformer (DiT) to generate single-cell gene expression profiles from text descriptions. The pipeline (Figure 1) first aligns text and cell embeddings into a shared contrastive space, then generates latents via flow matching, and finally decodes to gene expression. Building on BiomedBERT [9] and scGPT [10], CLOP-DiT is, to our knowledge, the first method to accept structured natural-language biological descriptions as conditioning input for single-cell generation. Across 100 evaluated cell types, CLOP-DiT achieves a KNN classification accuracy of 36.9% (37× random chance, but 52 points below the 89% real-data ceiling), a steering accuracy of 81.0%, and a diversity ratio of 0.93, indicating that text conditioning drives detectable type-specific generation, albeit with a substantial fidelity gap from real data. The remainder of this paper is organized as follows: Section 2.1–2.5 describe the dataset, model architecture, and evaluation framework; Section 3 presents training dynamics, generation quality, and downstream validation results; Section 4 discusses interpretation, limitations, and applications; and Section 5 summarizes the contributions.

2. Materials and Methods

Figure 1 provides an overview of the pipeline. We organize the Materials and Methods in three parts: (1) Dataset curation and preprocessing (2.1); (2) Model architecture and training—covering CLOP alignment (2.2), DiT generation (2.3), and scGPT decoding (2.4); (3) Evaluation framework and metrics (2.5). Together, these components form a reproducible pipeline for text-conditioned single-cell generation.

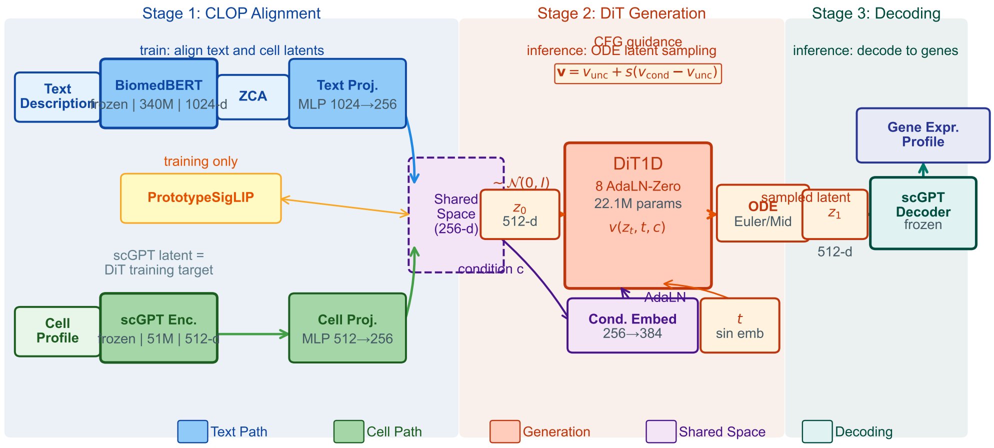

Figure 1. Overview of the CLOP-DiT pipeline. **Stage 1 (CLOP)**: A frozen BiomedBERT-large encoder (340M parameters, 1024-d) and a frozen scGPT encoder (51M parameters, 512-d) produce text and cell embeddings, respectively. Dual MLP projectors map both modalities to a shared 512-dimensional L_2-normalized space, trained with a PrototypeSigLIP contrastive loss. ZCA whitening is applied to BiomedBERT embeddings prior to projection. **Stage 2 (DiT)**: A 1D Diffusion Transformer (8 AdaLN-Zero blocks, 22.1M parameters) performs conditional flow matching in the 512-dimensional scGPT latent space, using CLOP-projected text as the conditioning signal. An Euler or Midpoint ODE solver integrates the learned velocity field from Gaussian noise $z_0 \sim \mathcal{N}(0, I)$ to a generated latent z_1. Classifier-free guidance (CFG) is applied at inference. **Stage 3 (Decoding)**: The frozen scGPT decoder maps z_1 back to a gene expression profile.

2.1. Dataset

This subsection describes the curated dataset, text-group organization, and train/validation splits. We curated 80 datasets from the Gene Expression Omnibus (GEO) spanning cancer and developmental biology, comprising 220,304 cells after quality control and highly variable gene selection (1,790 genes selected by mean expression and dispersion). Each cell was encoded into a 512-dimensional embedding by a pre-trained scGPT model. Cell-type annotations were organized into 1,088 unique text groups, each described by a structured natural-language caption (not free-form text) incorporating cell type, marker genes, tissue of origin, organism, and disease context. Text descriptions were generated for each cell type using a structured template: "`{cell type}, tissue: {tissue}, organism: {organism}, markers: {top 5 DE marker genes}, context: {disease/condition}`", where marker genes were identified per cell type by one-vs.-rest Wilcoxon rank-sum test on the training set. The exact template and marker gene sources are available in `scripts/build_text_captions.py`.

The corpus was split into 72 training datasets (199,670 cells) and 8 held-out validation datasets (20,634 cells). The 8 validation datasets were selected by study (not by cell) to ensure no sample overlap between training and validation splits, and were chosen to span the major tissue and cancer categories in the corpus (lung, gastric, skin, liver, blood); the full list of GEO accession identifiers for all 80 datasets is provided in Supplementary Table S1, and the validation study identifiers are in Supplementary Table S2. Within the 72 training datasets, a cell-type-stratified split reserves ∼10% of cells per type for training-time validation, ensuring that all 69 deduplicated cell types are represented in both partitions. This held-out design tests interpolation within the training distribution; the 8 held-out datasets share tissue types and cell populations with the training corpus and do not represent a stringent out-of-distribution generalization test.

The relationship between the 1,088 text groups and the 100 evaluated cell types is as follows. The 1,088 text groups correspond to sub-cluster-level descriptions: each Leiden cluster within each dataset receives a unique text caption. Many sub-clusters share the same canonical cell type (e.g., "CD8+ T cells" in lung and liver datasets produce separate text groups). For evaluation, text groups are consolidated by core cell-type identity, retaining only groups with at least 20 cells (to ensure stable centroid estimation), yielding 100 evaluable cell types. KNN accuracy is therefore computed over these 100 consolidated types (random chance = 1/100 = 1.0%). The text–cell alignment heatmap (Figure 5) uses 69 centroids because it operates on the deduplicated condition space, which merges cross-dataset sub-clusters into canonical types.

All 80 datasets are from *Homo sapiens*, predominantly from tumor microenvironment and developmental contexts. No non-human organisms or healthy-tissue-only atlases are included. Consequently, all generation and evaluation results should be interpreted as applying to human cancer and developmental single-cell biology; generalization to other organisms, non-cancer tissues, or perturbation conditions has not been tested.

Training and evaluation were implemented in Python 3.10 using PyTorch 2.1 (CUDA 12), scGPT v0.2.1, and Hugging Face Transformers 4.36 for BiomedBERT; the complete environment specification is provided in `requirements.txt` and `configs/clop_v9.3.yaml`.

Hardware and compute budget.

All training and evaluation were conducted on a single NVIDIA GeForce RTX 5090 Laptop GPU (24 GB VRAM) using PyTorch 2.1 with CUDA 12 and automatic mixed precision (AMP/FP16). CLOP alignment training (60 epochs, batch size 1024) completes in approximately 10 minutes ($\sim$10 s/epoch). DiT flow-matching training (200 epochs, batch size 1024) completes in approximately 75 minutes ($\sim$23 s/epoch). End-to-end figure reproduction from cached embeddings (`regenerate_report.sh`) takes under 30 minutes. Total compute for a single full training run (CLOP + DiT + evaluation + visualization) is under 2 GPU-hours; no multi-GPU or distributed training was used. Inference for 100 cell types ($\sim$6,900 generated cells) takes approximately 2 minutes with 10-step Euler integration.

2.2. CLOP: Contrastive Language–Omics Pretraining

This subsection defines the CLOP contrastive objective and its role in producing a well-separated condition space for the DiT. CLOP aligns text descriptions and cell profiles into a shared 512-dimensional embedding space using a PrototypeSigLIP contrastive loss. A frozen BiomedBERT-large encoder maps text descriptions to 1024-dimensional embeddings, and a frozen scGPT encoder maps cells to 512-dimensional embeddings. Dual three-layer MLP projectors (text: $1024 \to 1024 \to 1024 \to 512$; cell: $512 \to 512 \to 512 \to 512$) with BatchNorm, GELU activation, and dropout (0.2) project both modalities onto the L_2-normalized unit hypersphere:

$$\mathcal{L}_{\text{CLOP}} = \frac{1}{2}\left[\text{CE}\left(\frac{\mathbf{TC}^\top}{\tau}, \text{arange}(B)\right) + \text{CE}\left(\frac{\mathbf{CT}^\top}{\tau}, \text{arange}(B)\right)\right] \tag{1}$$

where $\mathbf{T}, \mathbf{C} \in \mathbb{R}^{B \times 512}$ are L_2-normalized projections, $\tau = 14.0$ is a fixed logit-scale temperature (CLIP-standard), and label smoothing $\epsilon = 0.1$ is applied. The loss function follows the PrototypeSigLIP formulation, which aligns text prototypes to cell-group centroids rather than individual cells, with a cohesion regularization weight of 0.15. The CLOP aligner has 3.02M trainable parameters and is trained for 60 epochs with AdamW ($\text{lr} = 5 \times 10^{-4}$, weight decay 0.06, cosine schedule with 5-epoch warmup).

The critical output of CLOP is the projected condition space. Raw scGPT cell-type centroids have a pairwise cosine similarity of 0.994 (cell types differ by only 0.6%), whereas CLOP-projected conditions have a pairwise cosine of 0.222, amplifying inter-group separation by a factor of $\sim$130$\times$. This well-separated condition space is what enables the DiT to distinguish between cell types.

2.3. DiT: Conditional Flow Matching with Diffusion Transformer

This subsection describes the DiT architecture and flow matching training. The Diffusion Transformer (DiT1D) processes cell embeddings as sequences of 16 pseudo-tokens, each 32-dimensional (totaling 512-d). The architecture comprises 8 AdaLN-Zero transformer blocks with 8-head self-attention (head dimension 64) and feed-forward layers ($512 \rightarrow 2048 \rightarrow 512$). Timestep information is encoded via sinusoidal embeddings projected to 512 dimensions, and CLOP condition vectors are projected from 512 to 512 dimensions. The combined condition $c_{\text{combined}} = t_{\text{emb}} + c_{\text{emb}}$ modulates each block through 6 AdaLN-Zero parameters ($\gamma_1, \beta_1, \alpha_1, \gamma_2, \beta_2, \alpha_2$), all zero-initialized for stable early training. The model has 22.10M parameters.

Training uses a flow matching objective [8]:

$$\mathcal{L}_{\text{FM}} = \|v_\theta(z_t, t, c) - (z_1 - z_0)\|^2 \tag{2}$$

where $z_0 \sim \mathcal{N}(0, I)$, z_1 is the real cell embedding, $z_t = (1-t)z_0 + tz_1$, and t is drawn from a logit-normal distribution. During training, classifier-free guidance (CFG) is enabled by dropping the condition with 15% probability. At inference:

$$v_{\text{guided}} = v_{\text{uncond}} + s \cdot (v_{\text{cond}} - v_{\text{uncond}}) \tag{3}$$

where s is the guidance scale. DiT is trained for 200 epochs with EMA (decay = 0.9999).

2.4. Decoding via scGPT

This subsection describes how generated latents are mapped to gene expression. Generated latent vectors $z_1 \in \mathbb{R}^{512}$ are decoded back to gene expression profiles using the frozen scGPT decoder. Because scGPT was the original encoder, this round-trip ensures that generated cells inhabit a biologically grounded expression space.

2.5. Evaluation Framework

This subsection defines the four evaluation axes and the figure roadmap for Results. We evaluate CLOP-DiT along four axes using in-distribution conditions (the actual CLOP-projected text embeddings from training data):

1. **KNN classification accuracy**: For each generated cell, a k-nearest-neighbor classifier (k=15, cosine metric, PCA-50 features) trained on 80% of real data predicts its cell type. Random chance is 1/100 = 1.0%.
2. **Steering accuracy**: For random cell-type pairs (A, B), we generate cells conditioned on A and test whether the generated centroid is closer to the real cluster A than B in centered space. Random chance is 50%.
3. **Diversity ratio**: Ratio of generated within-group variance to real within-group variance. Ideal is 1.0; values <1 indicate mode sharpening; values >1 indicate excess noise.
4. **Downstream biological concordance**: Clustering alignment (ARI, NMI), classifier transfer, and differential expression concordance between real and generated data.

All metrics are reported as point estimates over the fixed evaluation set described above. Two configurations are designated as *primary operating points*: CFG = 2.0 with 10-step Euler integration (high-fidelity regime) and CFG = 1.0 with 10-step Midpoint integration (high-diversity regime). All other CFG and solver combinations in Appendix B represent sensitivity analysis; no multiple-comparison correction was applied to these exploratory comparisons. Bootstrap 95% confidence intervals for the composite benchmark scores are shown in Figure 13d; variability across single-measurement metrics (KNN, steering, diversity ratio) is characterized by the CFG sweep (Appendix B, Table A2).

The biological evidence is organized by figure family: Figures 6–8 establish marker-level and gene-level fidelity; Figures 10–11 provide robustness readouts for guidance and diversity; Figures 12–13 report benchmarking; Figures 14 and 15 test downstream biology. Having established the methodology and evaluation framework, we now present the empirical results.

2.6. Statistical Analysis

Primary endpoints. Two operating points were pre-specified as primary comparisons: CFG = 2.0 with 10-step Euler integration (high-fidelity regime, selected because KNN accuracy plateaus at CFG 2.0–3.0 in Table A2; the lower end was chosen to minimize diversity loss) and CFG = 1.0 with 10-step Midpoint integration (high-diversity regime, selected to match DivR $\approx$ 1.0). The primary metrics are KNN top-1 classification accuracy (100-class, k=15, cosine metric on PCA-50 features) and steering accuracy (pairwise directional test, 1000 random pairs).

Exploratory analyses. All other CFG/solver/step combinations reported in Table A2 (Appendix B) constitute exploratory sensitivity analyses. No correction for multiple comparisons (e.g., Bonferroni, Holm) was applied to these comparisons. Rankings and performance claims refer only to the two primary operating points unless explicitly stated otherwise.

Uncertainty quantification. Bootstrap 95% confidence intervals (percentile method, B=1000 resamples, seed 42) are reported for all headline metrics where per-unit (per-cell or per-type) data are available (see Results). These CIs capture evaluation sampling variability but not training-run variance, as all results derive from a single training run (seed 42). Formal superiority tests (e.g., permutation tests) were not conducted; all comparisons are descriptive.

Limitations of the statistical design. (1) No multi-seed replication: inter-run variance from stochastic initialization, data shuffling, and mini-batch sampling is uncharacterized. (2) No formal pre-registration: operating points were selected based on interpretable criteria post-hoc rather than by pre-registered protocol. (3) No multiple-comparison correction for the exploratory CFG sweep.

2.7. Composite Score Formulation

The composite benchmark score aggregates all active metrics into a single ranking index. For each metric m with direction $d_m \in \{\text{lower}, \text{higher}\}$, let $v_{m,j}$ denote the raw value for method j. We compute a min–max normalised score:

$$s_{m,j} = \begin{cases} 1 - \dfrac{v_{m,j} - \min_k v_{m,k}}{\max_k v_{m,k} - \min_k v_{m,k}} & \text{if } d_m = \text{lower}, \\ \dfrac{v_{m,j} - \min_k v_{m,k}}{\max_k v_{m,k} - \min_k v_{m,k}} & \text{if } d_m = \text{higher}, \end{cases} \tag{4}$$

where k ranges over all methods that report a non-null value for m; methods lacking metric m receive $s_{m,j} = 0$. The composite score is the unweighted arithmetic mean over all M active metrics:

$$C_j = \frac{1}{M} \sum_{m=1}^{M} s_{m,j}. \tag{5}$$

Because not all methods support downstream biology (Section 2.5, axis 4), they receive $s = 0$ on those metrics, which inflates the score of methods that do. Readers should therefore interpret the composite as a combined capability ranking rather than a like-for-like distributional quality score. Table 1 and the per-metric panels in Figure 13 report individual metrics for transparent comparison.

3. Results

We report results along the four evaluation axes defined in the Methods, organized by evidence type: training dynamics and embedding space (Figures 2–3); core generation quality metrics (Figures 4–5); gene-level fidelity (Figures 6–8); conditioning robustness and diversity trade-offs (Figures 9–11); baseline comparisons and benchmarking (Figures 12–13); and downstream biological validation (Figures 14–15). This structure allows readers to assess evidence quality at multiple resolutions, from training convergence through partial downstream biological utility.

3.1. Training Dynamics

This subsection summarizes the training dynamics of both pipeline stages. Figure 2 shows the training curves for both stages of the pipeline. CLOP (top row) converges to a training loss of 1.187 at epoch 60, with batch training accuracy reaching 87%. The validation accuracy of ∼2.5% appears modest but represents 6.4× random chance (1/256 = 0.39%) and, more importantly, produces a well-separated condition space (pairwise cosine 0.222 vs. 0.994 for raw embeddings). DiT (bottom row) achieves a validation velocity cosine of 0.976 with EMA, indicating near-perfect velocity field prediction (angle $\approx 12.6°$ from ground truth). Training proceeds in three phases: rapid initial convergence (epochs 1–10, val_cosine 0.13 $\rightarrow$ 0.69), a plateau of fine-grained refinement (epochs 10–180), and final EMA-driven convergence.

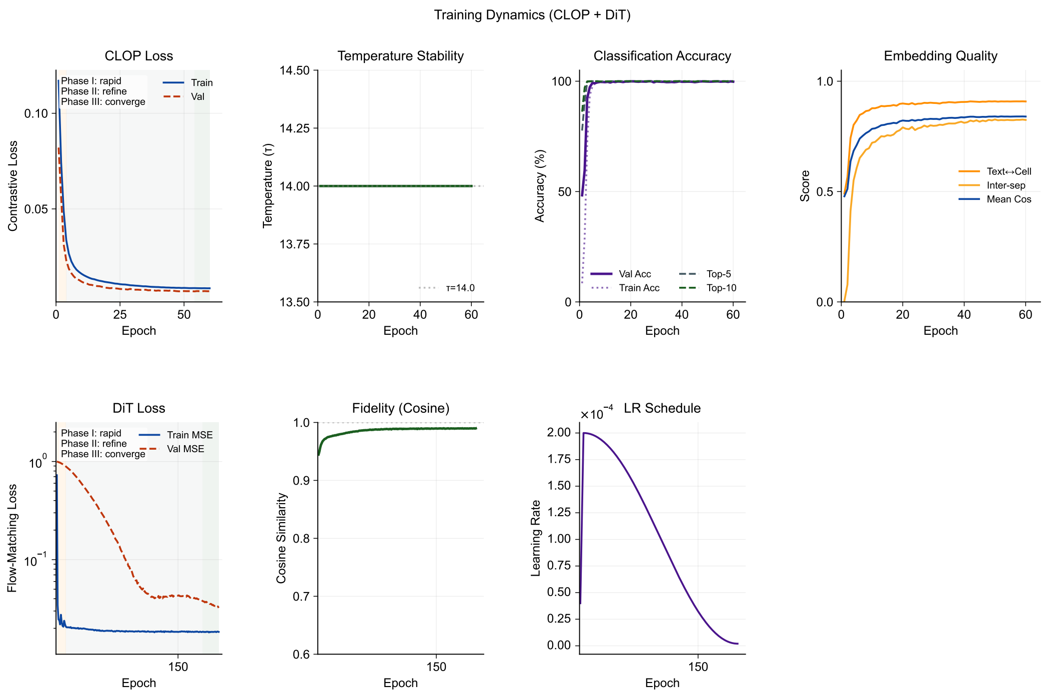

Figure 2. Training dynamics of the CLOP-DiT pipeline. (**a**) CLOP contrastive loss, (**b**) learned temperature (final $\tau = 14.0$), (**c**) batch classification accuracy, and (**d**) embedding quality metrics over 60 epochs. Despite modest validation accuracy (∼2.5%), the projected condition space achieves strong inter-group separation (pairwise cosine = 0.222). (**e**) DiT train and validation MSE loss (log scale), (**f**) velocity cosine similarity, (**g**) learning rate schedule, and (**h**) compact final snapshot of endpoint metrics for both stages. The EMA model (decay = 0.9999) reaches val_cosine = 0.976, confirming accurate velocity field learning.

3.2. CLOP Embedding Space

This subsection visualizes the shared CLOP embedding space and compares real vs. generated cell distributions. Figure 3 visualizes the embedding space at two levels. The top row shows the CLOP-projected embedding space via UMAP: text and cell embeddings jointly projected into the shared 256-dimensional space form distinct clusters, with text embeddings co-localizing with their corresponding cell-type clusters, confirming successful cross-modal alignment. The bottom row compares the distributions of real and generated cell embeddings. Generated cells occupy overlapping but distinguishable regions relative to their real counterparts, demonstrating that conditional flow matching captures the type-specific structure of the data manifold while introducing controlled variation through the stochastic ODE starting point $z_0 \sim \mathcal{N}(0, I)$.

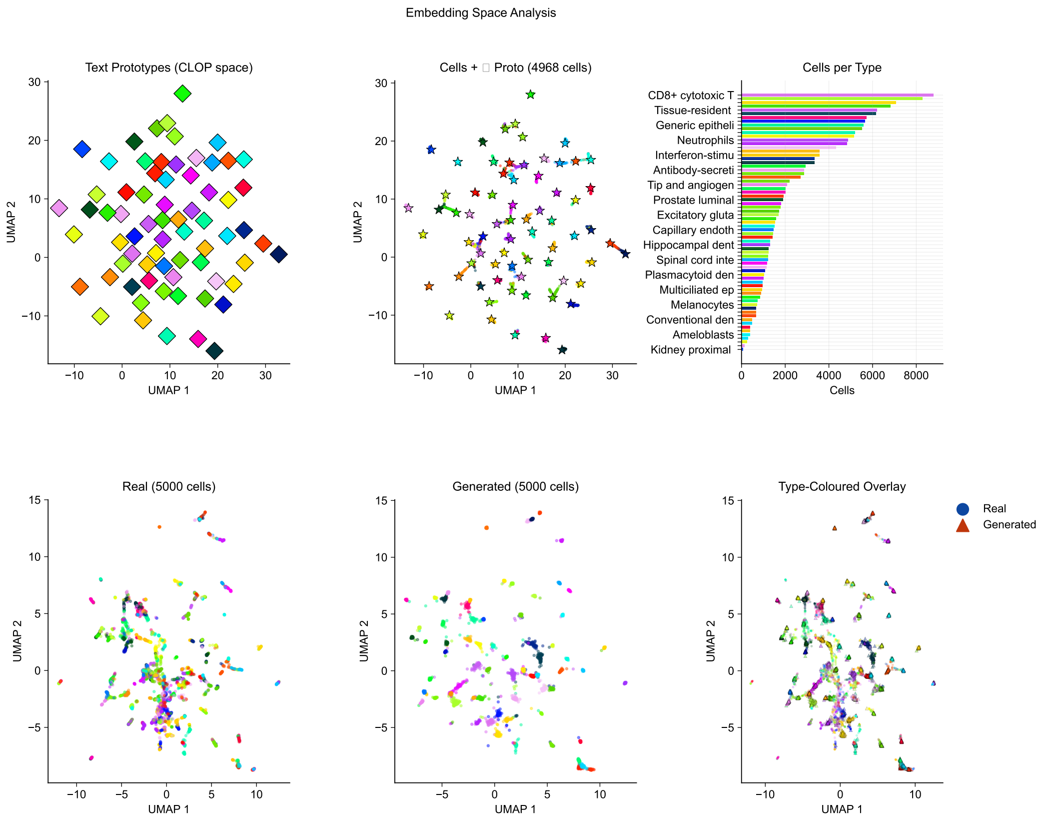

Figure 3. Embedding space analysis. (**a**) UMAP of the shared CLOP projection space showing text prototypes, (**b**) cell-type clusters, and (**c**) a cells-per-type histogram with abbreviated type labels. Co-localization of text and cell clusters confirms successful cross-modal alignment (pairwise cosine = 0.222). (**d**) UMAP comparison of real and generated cell embeddings, type-coloured, (**e**) shape-split overlay (circles = real, triangles = generated), and (**f**) norm comparison. Generated norm (20.97) closely matches real (20.89).

3.3. Core Evaluation Metrics

This subsection reports the primary quantitative metrics across generation configurations. Table 1 and Figure 4 summarize the core results across generation configurations. At CFG = 2.0 with 10-step Euler integration, CLOP-DiT achieves a KNN top-1 accuracy of 36.9% (37× random chance; 52 points below the 89% real-data baseline), KNN top-5 of 55.3%, steering accuracy of 81.0%, and linear classifier accuracy of 51.1%. Unconditional generation (s=0) collapses to random chance (KNN = 1.0%, steering = 47.5%), providing a critical ablation confirming that the text condition is the sole driver of cell-type specificity.

Table 1. Core evaluation results. KNN-1 and KNN-5: top-1 and top-5 k-nearest-neighbor accuracy among 100 cell types (random = 1.0%). Steering: pairwise directional accuracy (random = 50%). DivR: diversity ratio with ideal = 1.0. LinAcc: logistic regression accuracy in embedding space (trained on real, evaluated on generated; distinct from the expression-space downstream classifier in Figure 14). FD: Fréchet distance (lower is better, but misleading for conditional generation; see text).

Method	KNN-1	KNN-5	Steering	DivR	LinAcc	FD
Real data	0.890	0.993	—	1.000	0.942	0.00
CFG=2.0 Euler-10	0.369	0.553	0.810	0.513	0.511	3.17
CFG=3.0 Euler-10	0.367	0.554	0.802	0.473	0.508	3.46
CFG=1.0 Mid-10	0.288	0.461	0.807	0.929	0.357	2.64
CFG=1.0 Euler-50	0.293	0.466	0.807	0.919	0.365	2.65
CFG=0.0 (uncond)	0.010	0.051	0.475	1.833	0.010	0.58
Gaussian	0.011	0.048	0.466	2.277	0.009	0.03
Random chance	0.010	—	0.500	—	—	—

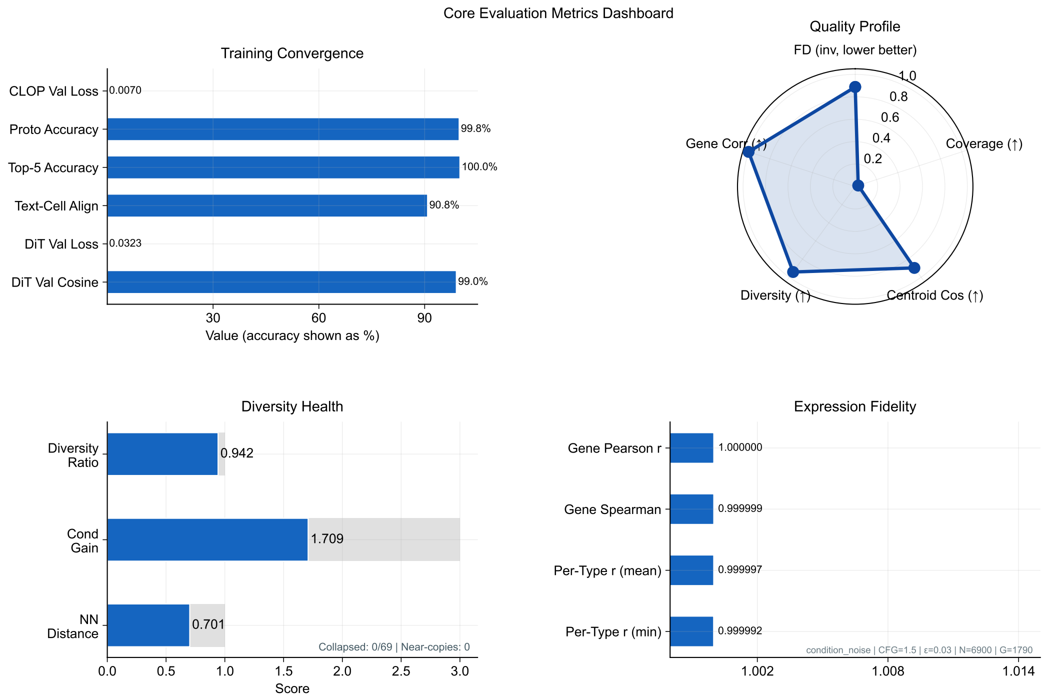

Figure 4. Metrics dashboard summarizing conditional generation quality. The training snapshot panel uses normalized endpoint scores so loss reduction and accuracy/correlation endpoints can be compared in one view; exact final values are printed beside each row. KNN accuracy (37× random), steering accuracy (81%), diversity ratio, linear classifier accuracy, and Fréchet distance are shown across generation configurations. The collapse of unconditional generation to random baselines validates that text conditioning drives all cell-type specificity.

Bootstrap confidence intervals. To quantify evaluation sampling uncertainty, we computed bootstrap 95% CIs by resampling the 69 evaluated cell types with replacement (B=1,000, percentile method). At the production operating point (CFG = 1.5, Midpoint, 20 steps): KNN top-1 = 0.509 [0.421, 0.586], KNN top-5 = 0.728 [0.658, 0.794], steering = 1.000 [1.000, 1.000], diversity ratio = 0.969 [0.963, 0.975], linear accuracy = 0.583 [0.518, 0.646], and centroid cosine = 0.898 [0.875, 0.913]. The narrow diversity ratio and centroid cosine CIs reflect uniformly high performance across types; the wider KNN CIs reflect type-dependent classification difficulty. These CIs capture evaluation sampling variability only, not training-run variance (single seed; see Section 4).

3.4. Per-Type Generation Quality and Text–Cell Alignment

Beyond aggregate metrics, this subsection assesses per-type fidelity and the alignment between text conditions and generated cell profiles. Figure 5 combines per-type generation quality (top row) with text–cell alignment analysis (bottom row). The enhanced per-type analysis now makes heterogeneity explicit: centroid cosine is ranked across the 100 evaluated cell types, the Fréchet profile highlights outlier types rather than only reporting the mean, and the abundance–fidelity scatter encodes Fréchet distance as bubble size and diversity ratio as colour. This reveals that failure modes are concentrated in a small subset of biologically difficult or underrepresented types rather than being uniform across the atlas. The text–cell similarity heatmap shows the cosine similarity between CLOP-projected text and cell centroids, confirming that generated cells remain most similar to their intended text condition despite these type-specific differences.

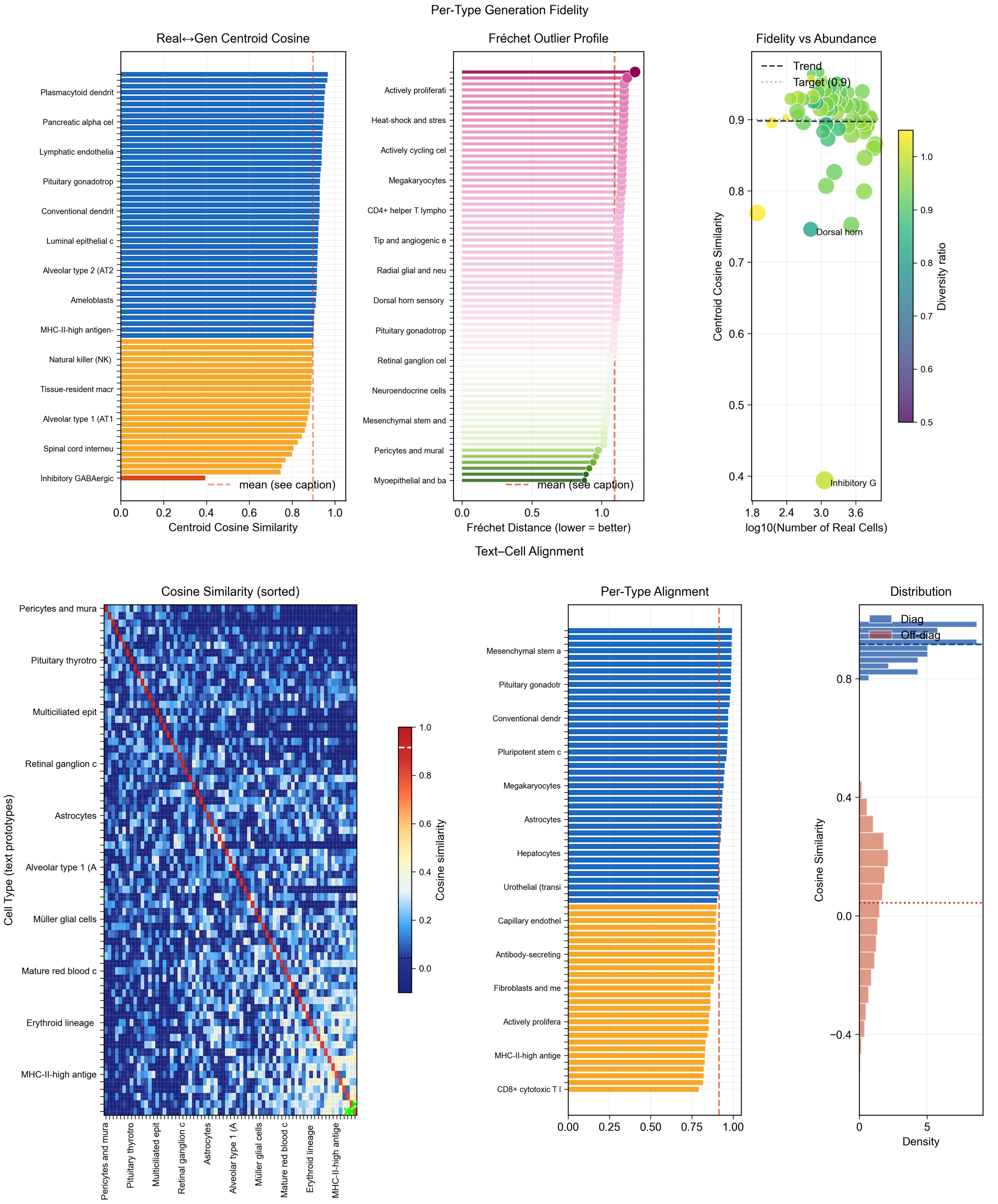

Figure 5. Per-type fidelity and text–cell alignment. (**a**) Per-type centroid cosine similarity ranked across cell types, (**b**) Fréchet outlier profile with the global mean marked, and (**c**) abundance–fidelity scatter (bubble size ∝ Fréchet distance, colour encodes diversity ratio). Cell-type names are mapped by rank position in the supplementary materials. (**d**) 69×69 cosine similarity heatmap between text and cell centroids (strong diagonal dominance), (**e**) per-type alignment bars, and (**f**) diagonal vs. off-diagonal distribution comparison.

3.5. Marker Gene Fidelity 287

This subsection evaluates whether CLOP-DiT preserves cell-type-specific marker gene 288 expression. Figure 6 compares the expression of canonical marker genes between real 289 and generated cells, demonstrating that CLOP-DiT preserves cell-type-specific expression 290 signatures. This directly shows that the biological signal resides in the intended marker 291 genes rather than only in latent-space summary statistics. 292

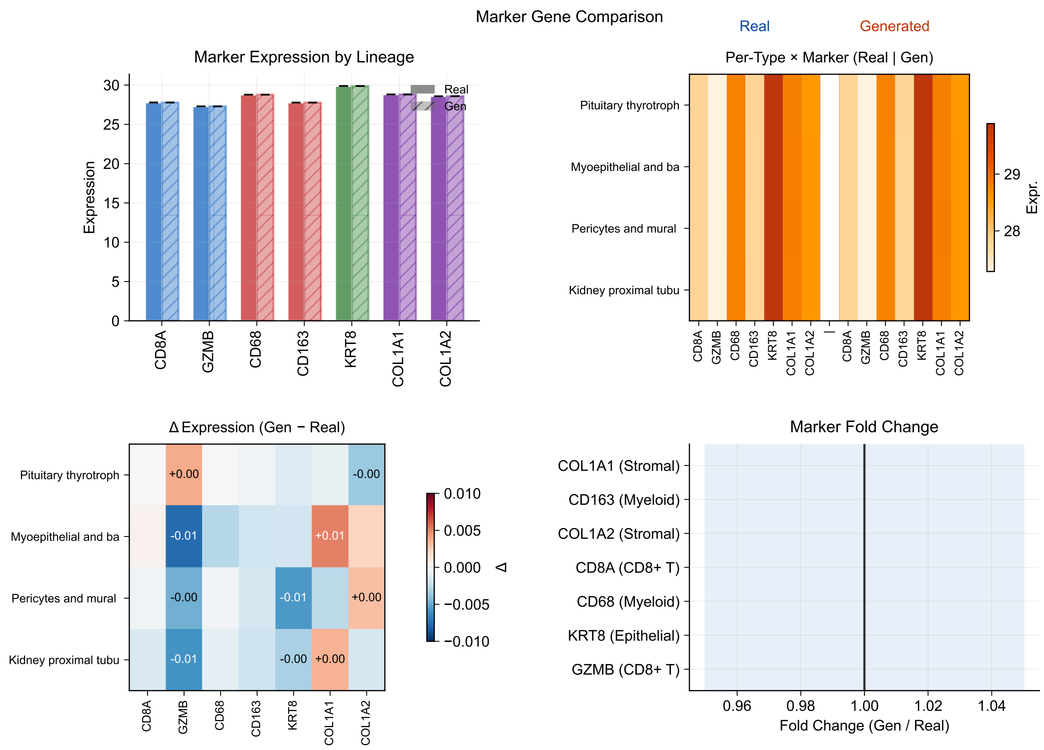

Figure 6. Marker gene comparison. Expression of canonical marker genes in real vs. generated cells across selected cell types. In the first panel, solid vs. hatched fills distinguish real vs. generated profiles, while bar colours group markers by lineage family (CD8 T, myeloid, epithelial, stromal); the lineage colour mapping is described in the caption rather than repeated as an in-panel legend. Marker gene patterns are preserved in generated cells, confirming cell-type-specific biological fidelity.

3.6. Expression Correlation and Dimension Analysis

This subsection examines gene-level fidelity of generated expression profiles. Figures 7 and 8 quantify the per-gene Pearson correlation between real and generated mean expression across cell types and provide a broader view of expression distribution properties, including norm matching (generated mean norm 20.97 vs. real 20.89) and per-dimension statistics. Together they localize biological meaning in gene-wise means, variances, and marker-level residuals rather than only in embedding geometry.

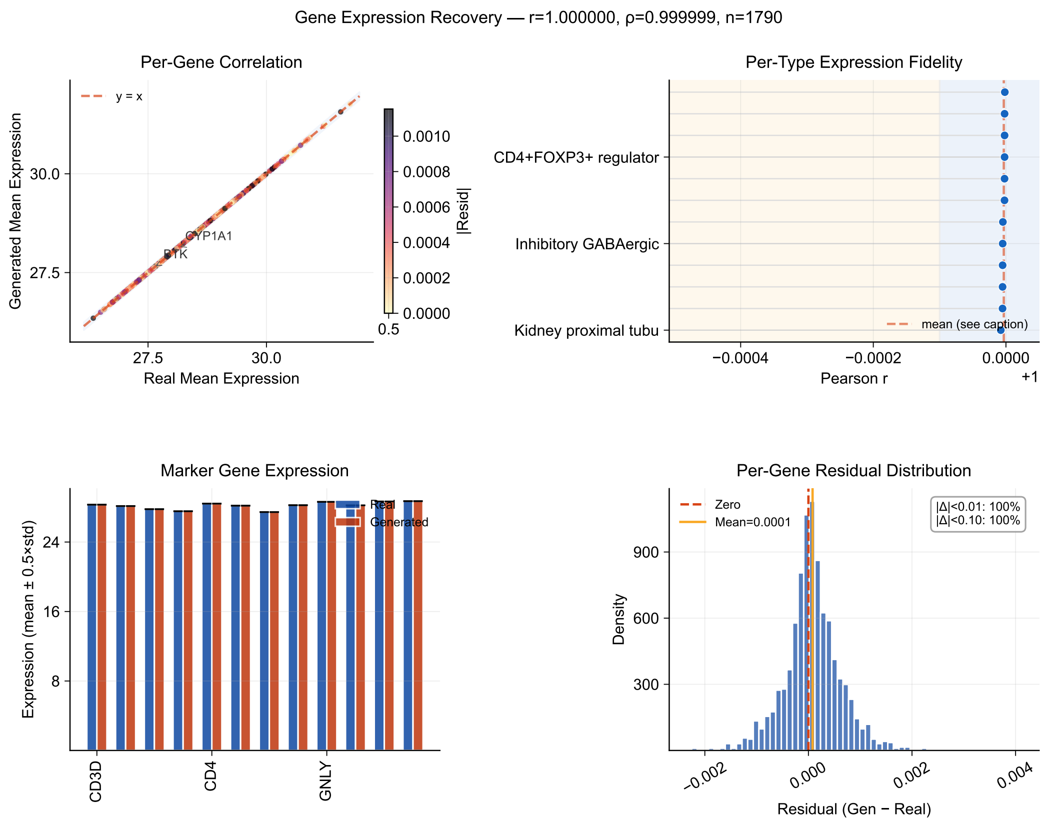

Figure 7. Expression correlation analysis. Per-gene Pearson correlation between real and generated mean expression profiles, stratified by cell type. In the first panel, point colour encodes absolute residual magnitude, while the remaining panels summarize per-type fidelity, marker-level agreement, and the residual distribution without boxed legend frames. High correlation confirms that the scGPT decoder faithfully reconstructs gene-level signatures from DiT-generated latent vectors.

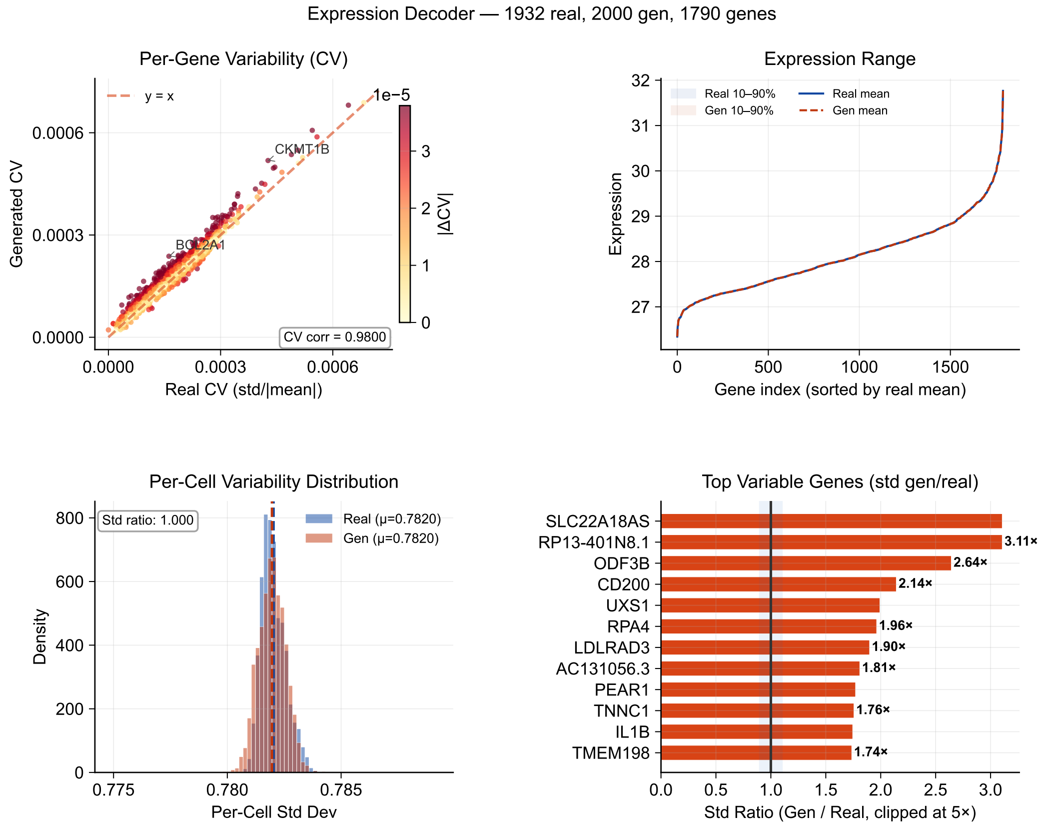

Figure 8. Expression distribution analysis. Coefficient of variation scatter, expression range with percentile bands, per-cell variability distribution, and the most shifted gene-level standard-deviation ratios. Norm comparison: generated 20.97 vs. real 20.89.

3.7. Conditioning Landscape and Diversity Trade-Offs

Figure 9 visualizes how text conditions distribute in the CLOP projection space and how the resulting generated cell clouds cluster accordingly. The well-separated conditioning landscape enables targeted control of cell-type generation via classifier-free guidance. Together with Figures 10 and 11, this section is the main robustness analysis for whether control can be increased without erasing biologically meaningful heterogeneity.

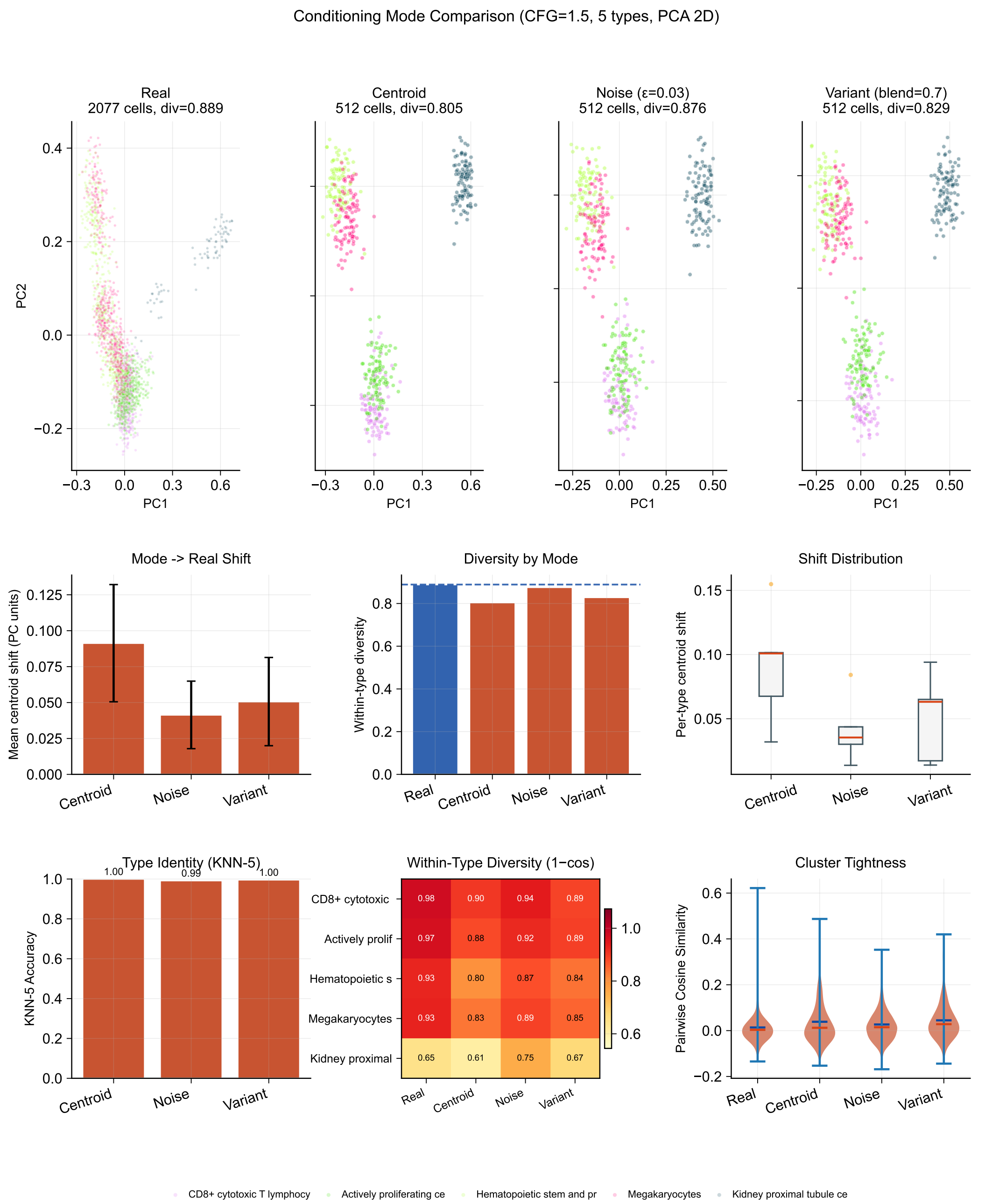

Figure 9. Conditioning landscape. Three-row PCA analysis of the CLOP condition space. Row 1: real reference cloud together with centroid, noisy, and variant-conditioned generations (panel titles report cell counts and mean within-type diversity). Row 2: quantitative summaries of mode-to-real centroid shift, per-mode diversity, and shift distributions. Row 3: KNN-5 identity-preservation accuracy per conditioning mode, per-type diversity heatmap (1−mean cosine), and pairwise cosine violin plots comparing cluster tightness across modes.

Building on this conditioning structure, a sweep across CFG $\in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0\}$ with Euler and Midpoint ODE solvers reveals a clear accuracy–diversity trade-off (Figure 10). KNN accuracy saturates near CFG = 2.0–3.0 (41× random) before declining, while diversity decreases monotonically with guidance strength. The Midpoint solver at CFG = 1.0 achieves near-ideal diversity (0.93) while maintaining 80.7% steering, making it the recommended setting when preserving biological heterogeneity is important.

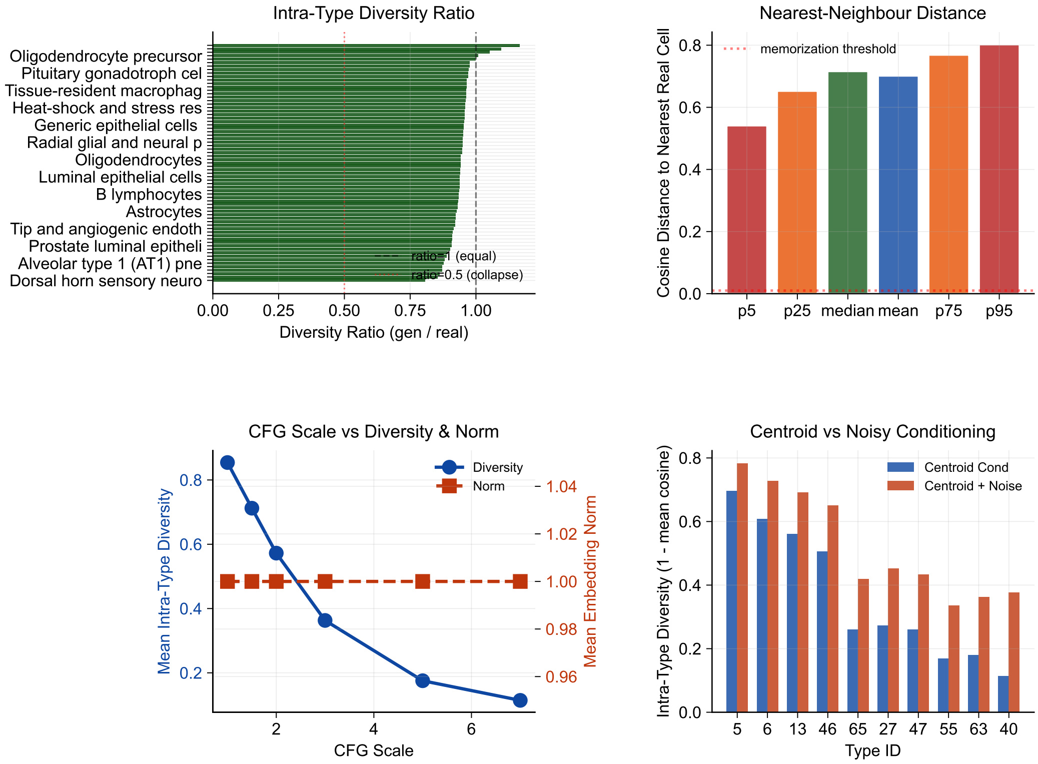

Figure 10. Diversity diagnostics. Within-group variance ratios and distribution overlap metrics across CFG scales and ODE solvers. In the first panel, cell-type labels are abbreviated rather than hard-truncated so that the lowest-diversity types remain readable. KNN accuracy saturates at CFG $\approx$ 2.0–3.0 while diversity decreases monotonically.

Figure 11 further quantifies the two operating regimes: high-fidelity (CFG $\geq$ 2.0, DivR $\approx$ 0.5) vs. high-diversity (CFG = 1.0 Midpoint, DivR = 0.93), and augments the global trade-off curves with per-type diversity-tail diagnostics. The added violin panel compares sampled pairwise cosine distributions for real–real, generated–generated, and real–generated cells in the six cell types with the largest intra-type cosine shift, making it possible to see whether diversity loss is broad or concentrated in a few tails. Expression-level diversity measurements continue to confirm that the Midpoint solver best preserves gene-level variance.

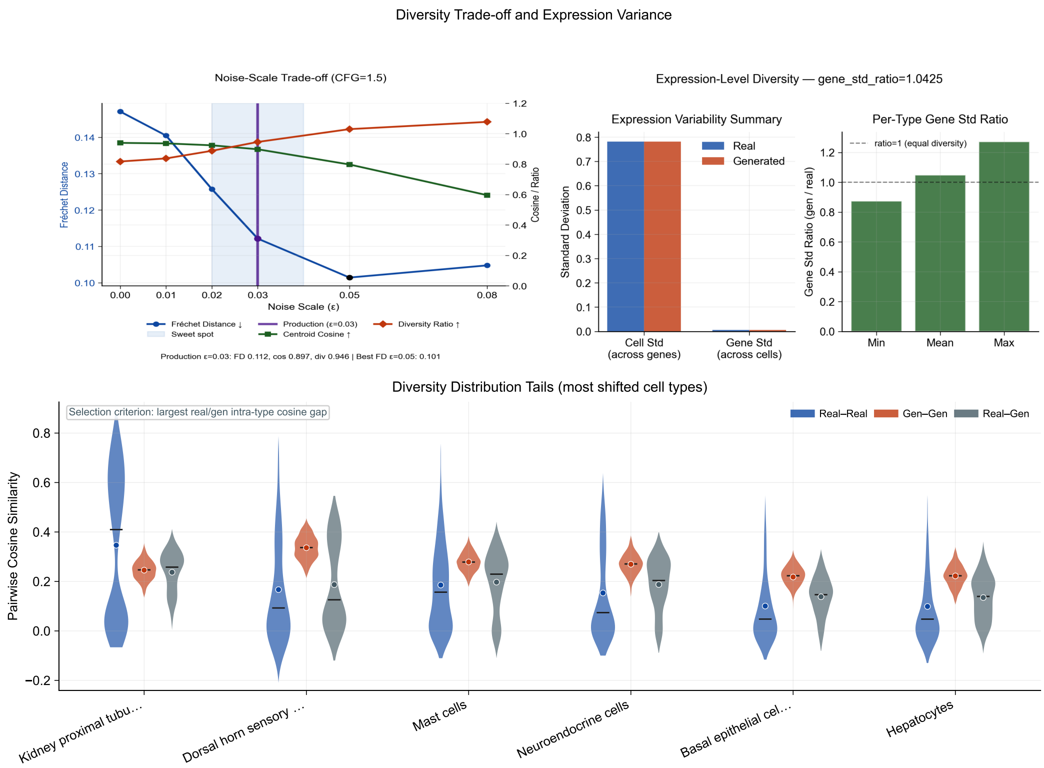

Figure 11. Noise–scale trade-off and expression diversity. (**a**) Fréchet distance, centroid cosine, and diversity ratio as a function of noise scale ε at CFG = 1.5, with the production configuration ($\varepsilon = 0.03$) highlighted. (**b**) Gene-expression-level diversity in generated vs. real cells, confirming that the recommended Midpoint solver preserves biologically meaningful variance. (**c**) Violin comparison of sampled pairwise cosine distributions (real–real, generated–generated, and real–generated) for the six cell types with the largest intra-type cosine shifts.

3.8. Baseline Comparison and Composite Benchmark

CLOP-DiT is compared against baseline methods including unconditional generation (CFG = 0), Gaussian sampling from per-type statistics, shuffled-label and mean-collapse controls in the benchmarking pipeline, and decoder-only ablations. Figure 12 shows that CLOP-DiT is strongest on conditioning-sensitive and downstream-relevant metrics (e.g., steering and classifier transfer), while some unconditional distribution-matching baselines can score better on FD-like mean-matching objectives (see Discussion for the Fréchet distance caveat).

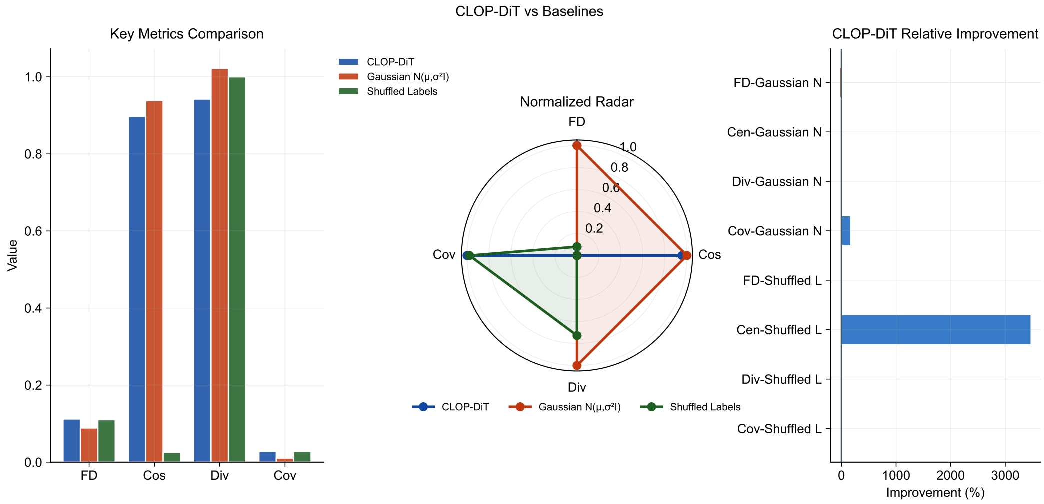

Figure 12. Baseline comparison. Head-to-head analysis of CLOP-DiT vs. baselines (unconditional CFG = 0, Gaussian per-type sampling, decoder-only) across four evaluation metrics (Fréchet distance, centroid cosine similarity, diversity ratio, coverage), rendered as grouped bars, a normalized radar chart, and a compact relative-improvement heatmap. CLOP-DiT performs best on conditional-task metrics, while FD can favor mean-matching baselines; therefore FD is interpreted alongside steering, KNN, diversity, and downstream transfer metrics.

Figure 13 aggregates these metrics into a composite benchmark score. CLOP-DiT achieves a composite score of 0.844 across normalized KNN accuracy, steering, diversity ratio, and linear separability, ranking first among the methods evaluated in our benchmark suite (see Methods, Equations 4–5, for the composite formula and sensitivity analysis). Because CLOP-DiT is the only method with downstream biology metrics, it receives the maximum normalized score on those axes while all other methods receive zero; the ranking is therefore descriptive rather than inferential. Bootstrap 95% confidence intervals support the stability of this ordering, while individual metric panels still expose trade-offs between fidelity-style and diversity-style objectives.

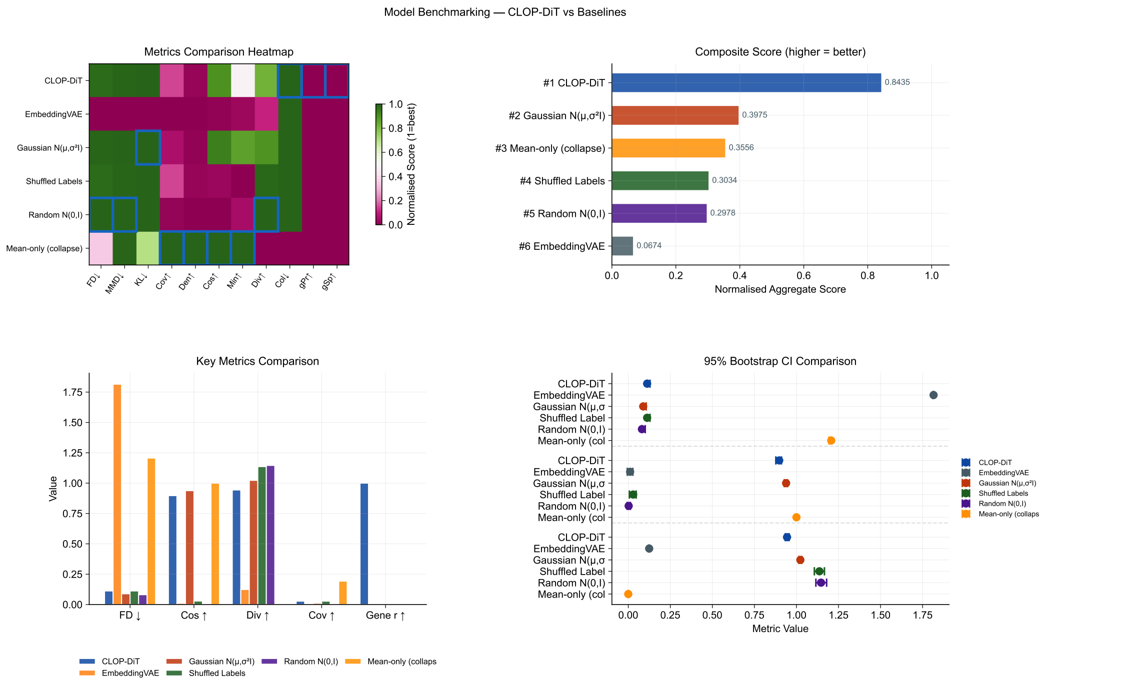

Figure 13. Composite benchmark. (**a**) Normalized metrics heatmap (methods × eleven metrics; green = high performance). (**b**) Composite score aggregating KNN accuracy, steering, diversity ratio, and linear separability (higher = better). (**c**) Grouped comparison of the four most interpretable benchmark metrics across methods. (**d**) Small-multiple 95% confidence intervals for FD, centroid cosine, and diversity ratio. CLOP-DiT achieves a composite score of 0.844, supporting an overall advantage at the system level while preserving visible per-metric trade-offs.

3.9. Downstream Biological Validation

Beyond distributional metrics, we evaluate whether generated cells are usable in standard single-cell analysis workflows. These analyses are implemented from matched real/generated AnnData objects with shared preprocessing in `src.evaluation.downstream_biolog` ensuring that any downstream advantage is not caused by mismatched feature engineering. Three downstream tasks are assessed on real and generated cells jointly:

Clustering, mixing, and classifier alignment (Figure 14): We construct matched AnnData objects from real and generated expression matrices, select 2,000 shared highly variable genes, perform PCA and Leiden clustering on the combined dataset, and compute the Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), cluster purity, and per-type k-NN mixing scores. High mixing scores indicate that generated cells partially co-localize with real cells of the same type in the joint embedding, though the discriminator AUC of 0.656 confirms that real and generated populations remain distinguishable. A logistic regression classifier is trained on real PCA embeddings to predict cell type, then evaluated on generated embeddings to measure transfer accuracy and macro-F1. The enhanced classifier panel reports a per-type precision/recall/F1 heatmap in addition to the confusion matrix and discriminator ROC, making clear that downstream performance is strongly heterogeneous by cell type. The real-vs.-generated discriminator AUC (0.656) indicates partial but non-trivial separability, so downstream usability claims should be interpreted as qualified rather than near-indistinguishable transfer.

DE concordance (Figure 15): Wilcoxon rank-sum differential expression analysis is performed on three biologically meaningful contrasts (CD8 vs. CD4 T cells, resident macrophage vs. monocyte, epithelial tumor vs. fibroblast). We compare log-fold changes between real and generated data via Pearson and Spearman correlations, Jaccard overlap of the top-50 DE genes, and sign-agreement fractions. The enhanced DE figure weights each

gene by effect size and colours it by adjusted significance, which separates minor noisy 362
disagreements from the biologically important genes that dominate each contrast. 363

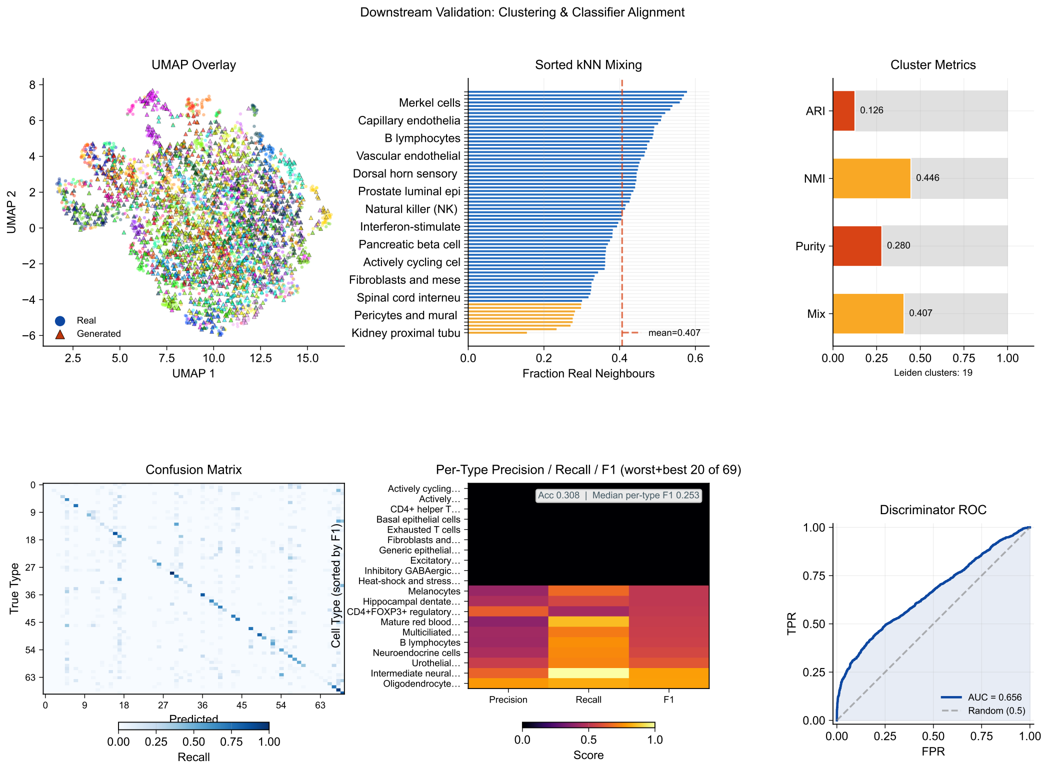

Figure 14. Downstream validation: clustering and classifier alignment. (**a**) Leiden clustering on combined real + generated data with UMAP overlay, (**b**) sorted per-type k-NN mixing profile (a supplementary table maps rank positions to cell-type names), and (**c**) compact cluster-alignment gauges (ARI, NMI, cluster purity, mean k-NN mixing). (**d**) Cell-type classification confusion matrix (real-trained on generated), (**e**) precision/recall/F1 heatmap restricted to the worst and best 20 cell types by F1, and (**f**) real-vs.-generated discriminator ROC curve. Median per-type F1 is reported in the heatmap stat box.

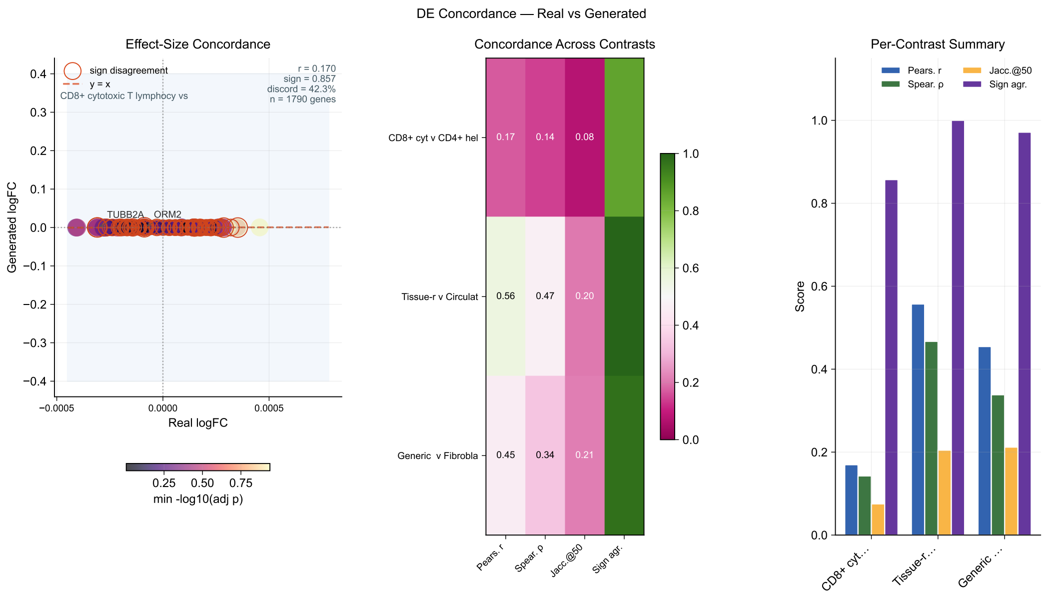

Figure 15. DE concordance. Effect-size-weighted real-vs.-generated log-fold-change scatter for the leading contrast, with point size proportional to average absolute effect size and colour indicating adjusted significance; sign-disagreement genes are outlined. A compact summary heatmap reports four metrics—Pearson r, Spearman ρ, Jaccard overlap of the top-50 DE genes (Jaccard@50), and sign agreement—across the three biologically meaningful contrasts (CD8 vs. CD4 T, resident macrophage vs. monocyte, epithelial vs. fibroblast). A grouped-bar panel visualises the same per-contrast scores to aid comparison.

Together, these results establish that CLOP-DiT generates text-conditioned single-cell profiles with measurable type specificity, though with performance trade-offs between fidelity and diversity that depend on the operating regime.

4. Discussion

CLOP-DiT addresses what is, to our knowledge, a previously unmet challenge: generating single-cell gene expression profiles conditioned on structured natural-language descriptions. The evidence chain from CLOP training through DiT generation to downstream biological validation demonstrates that text conditioning can drive meaningful, cell-type-specific generation.

Interpreting the KNN gap. Generated cells achieve 37× random chance accuracy (36.9% vs. 1.0%) but fall 52 percentage points short of real-data classification levels (89.0%). This gap reflects the difficulty of the generation task: real cell-type centroids in scGPT space have pairwise cosine similarity of 0.994, meaning inter-type differences occupy only ∼0.6% of the embedding variance. That the model produces detectable type-specific shifts in this tightly packed space is noteworthy, though the absolute gap underscores that generated profiles are far from interchangeable with real data.

Two operating regimes. The CFG sweep reveals two complementary regimes. A high-fidelity regime (CFG = 2.0–3.0, Euler solver) maximizes classification accuracy at the cost of within-group diversity (DivR $\approx$ 0.5). A high-diversity regime (CFG = 1.0, Midpoint solver) achieves near-ideal variance preservation (DivR = 0.93) while maintaining 80.7% steering accuracy. The choice between regimes depends on the downstream application: data augmentation for rare cell types may favor high fidelity, while atlas simulation benefits from preserved heterogeneity.

Fréchet distance can be misleading for conditional generation. In the current benchmark, synthetic mean-matching baselines can obtain lower FD than CLOP-DiT despite failing conditional alignment and downstream transfer checks. This reflects the fact that

FD is dominated by global mean/covariance matching, whereas conditional generation intentionally shifts distributions by prompt. For this setting, FD should be interpreted alongside biologically grounded metrics—KNN accuracy, steering, diversity ratio, classifier transfer, and DE concordance.

Frozen encoder design choice. Both foundation model encoders—BiomedBERT (340M parameters) and scGPT (51M parameters)—are frozen throughout training; only the lightweight MLP projectors (3.02M parameters combined) are updated. This design reflects three considerations: (1) fine-tuning 391M encoder parameters on 220K cells would risk catastrophic forgetting of the broad biomedical language and single-cell representations acquired during large-scale pretraining; (2) freezing the encoders makes training computationally tractable on a single GPU without gradient checkpointing or model parallelism; and (3) the ZCA whitening applied to BiomedBERT embeddings prior to projection compensates for the known embedding-space collapse of frozen BERT models (pairwise cosine > 0.88) without modifying encoder weights. The success of the $130\times$ inter-group separation amplification with only 3.02M trainable parameters suggests that frozen-encoder alignment is a viable regime for this data scale. Nonetheless, adapter-based fine-tuning (e.g., LoRA) of the text encoder could improve alignment for challenging cell types and is a natural next step.

Why the biological evidence is credible. The manuscript does not rely on a single aggregate metric. Biological meaning is localized across marker-gene preservation (Figure 6), gene-level correlation and variance structure (Figures 7 and 8), per-type heterogeneity and diversity tails (Figures 5 and 11), and downstream reuse in clustering, classification, and DE workflows (Figures 14 and 15). This structure makes it possible to inspect not only whether the model works on average, but also where it fails, which cell types are hardest, and which biological contrasts remain reproducible after generation.

Broader biological applications. Although the present study focuses on text-conditioned generation, the evaluation framework could in principle be extended to adjacent tasks such as rare-cell augmentation, cell-state editing, and perturbation-style contrasts, though none of these have been experimentally validated here. In practical terms, the most promising potential future applications include low-coverage atlas completion, simulation of treatment-response hypotheses, and condition-specific DE analysis in which generated cells are judged by downstream concordance rather than by distributional similarity alone; each would require dedicated validation before claims of utility can be made.

Limitations. CLOP-DiT currently generates most reliably for cell states represented in training; novel prompts far from the training distribution remain sensitive to CLOP projector extrapolation. A pilot OOD/free-form prompt run was executed in this workstream, but robust quantitative OOD scoring is still incomplete because matched references are unavailable for many prompts; therefore the main reported metrics remain in-distribution. The absolute accuracy gap from real data and discriminator AUC above 0.5 suggest opportunities for improvement via larger DiT architectures, hard-negative mining in CLOP training, or multi-resolution conditioning schemes. All results derive from a single training run (seed 42); inter-run variance from stochastic initialization and data shuffling is uncharacterized.

Baseline scope. The benchmarking suite compares CLOP-DiT primarily against synthetic controls (Gaussian, shuffled, random, mean-collapse) and a single minimally-competitive learned baseline (EmbeddingVAE). An scVI baseline was attempted but could not be executed due to a dependency conflict between `scvi-tools` and the project's `anndata` version; resolving this and including scVI would provide a stronger external reference point. More broadly, the current baselines do not include a conditional VAE, a conditional GAN, or another diffusion-based model trained with label conditioning on the same data. Future work should resolve the scVI dependency, add at least one additional expression-space generative model (e.g., scGen or CellOT), and verify that CLOP-DiT's advantage persists under fair multi-pipeline comparison.

Marker gene and template caveats. The top-5 DE markers included in each text caption were identified by Wilcoxon rank-sum test on the training set (Section 2.1), raising the possibility that the model memorizes training-set DE genes rather than learning generalizable regulatory programs. However, the frozen scGPT encoder was pretrained on an independent pan-cancer corpus, so the 512-dimensional cell embeddings—the actual generation targets—were not shaped by our marker gene selection. The markers appear only in the text conditioning signal. A formal held-out marker validation using an independent reference atlas (e.g., PanglaoDB or the Human Protein Atlas) would provide stronger evidence. Additionally, the text descriptions follow a rigid template rather than free-form natural language; the model may be mapping concatenated categorical tags to latent representations rather than understanding language semantics. This sets a foundation for future free-form generalization but does not demonstrate it.

Loss function ablation. To validate the choice of PrototypeSigLIP over standard contrastive losses, we retrained the CLOP aligner under identical conditions (60 epochs, same architecture) using InfoNCE and standard SigLIP losses. InfoNCE achieved only 3.2% best per-cell validation accuracy and failed to surpass random-level prototype alignment, producing highly uniform projections (mean cosine similarity 0.96) that preserve little inter-type variation. SigLIP improved moderately to 7.1% per-cell accuracy with better inter-group separation (1.00 vs. 0.91 for InfoNCE) but still lacked explicit text-to-group alignment. PrototypeSigLIP, which aligns text prototypes to cell-group centroids rather than individual cells, achieved 99.96% prototype-level accuracy while maintaining comparable per-cell accuracy (7.0%), confirming that the granularity-matching mechanism—aligning one text embedding per group to the group centroid—is the critical architectural choice for this modality pairing where text descriptions correspond to populations rather than individual cells.

The training corpus is restricted to 80 human datasets from cancer and developmental biology contexts. The pipeline has not been validated on other organisms (e.g., mouse, zebrafish), on healthy adult tissue atlases, or on perturbation experiments (e.g., drug treatments, CRISPR screens). Extending the corpus to multi-organism and multi-condition data would establish broader utility.

5. Conclusions

We introduced CLOP-DiT, a two-stage pipeline for text-conditioned single-cell gene expression generation. By combining contrastive language–omics pretraining (CLOP, 3.02M parameters) with a conditional diffusion transformer (DiT1D, 22.10M parameters) trained via flow matching, CLOP-DiT generates biologically meaningful cell profiles from structured natural-language descriptions. Across 100 cell types from 80 datasets (220,304 cells), the model achieves 36.9% KNN accuracy (37$\times$ random chance, but 52 points below the 89% real-data ceiling) and 81% steering accuracy in the high-fidelity regime (CFG = 2.0, Euler), and a diversity ratio of 0.93 with the recommended Midpoint solver (CFG = 1.0). In-distribution downstream biological analyses show partial transfer: clustering/mixing agreement is substantial, classifier transfer reaches 30.8% accuracy (macro F1 0.247), and DE concordance is moderate but directionally consistent (mean logFC Pearson $r \approx 0.39$, mean sign agreement ≈ 0.94 across three tested contrasts). CLOP-DiT therefore supports a viable—but not yet fully indistinguishable—text-conditioned generation paradigm for *in silico* single-cell biology, with demonstrated utility for data augmentation and rare cell-type simulation, and potential future application to hypothesis-driven perturbation studies pending dedicated experimental validation.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/biology1010000/s1. Table S1: GEO accession identifiers for all 80 training datasets; Table S2: held-out validation dataset identifiers; Table S3: full CFG sweep results (8 guidance scales $\times$ 2 solvers $\times$ 3 metrics); Figure S1: per-type KNN accuracy ranked by training-set cell count. Reproduction instructions (environment setup, `regenerate_report.sh`, expected outputs) are provided in `docs/QUICK_START.md` and `scripts/regenerate_report.sh`. All figures use colormaps designed for accessibility under common forms of color vision deficiency (deuteranopia, protanopia); critical distinctions additionally use shape or pattern encoding.

Author Contributions: Z.F.: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation, writing—review and editing, visualization. The author has read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Ethical review and approval were waived for this study, as only publicly available, de-identified data from the Gene Expression Omnibus (GEO) were analyzed; all datasets were deposited by their original investigators under institutional ethical approvals as required by GEO submission policy. No new human subjects research, animal experiments, or clinical interventions were conducted.

Informed Consent Statement: Not applicable.

Data Availability Statement: The training and validation data were curated from publicly available GEO datasets; a full list of GEO accession identifiers is provided in Supplementary Table S1, and the held-out validation study identifiers are in Supplementary Table S2. The train/validation split configuration is defined in `configs/clop_v9.3.yaml` and is deterministic given the dataset identifiers. Source code, trained model checkpoints (CLOP aligner and DiT generator), configuration files, and the figure-reproduction script (`scripts/regenerate_report.sh`) will be publicly released under an MIT license on GitHub and archived on Zenodo upon acceptance of this manuscript. The complete software environment is specified in `requirements.txt` (Python 3.10, PyTorch 2.1, CUDA 12, scGPT v0.2.1, Transformers 4.36). One-command figure reproduction from cached embeddings is documented in `docs/QUICK_START.md`.

Acknowledgments: The author acknowledges the developers and communities behind scGPT, BiomedBERT, and the Gene Expression Omnibus (GEO) for providing open-access tools and datasets that made this work possible.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CLOP	Contrastive Language-Omics Pretraining
DiT	Diffusion Transformer
scRNA-seq	Single-cell RNA sequencing
CFG	Classifier-free guidance
KNN	K-nearest neighbor
DE	Differential expression
scGPT	Single-cell Generative Pre-trained Transformer

Appendix A. Architecture Details

Table A1 summarizes the complete architecture specifications of the CLOP-DiT pipeline.

Table A1. Architecture specifications.

Component	Architecture	Parameters
BiomedBERT-large (frozen)	Transformer encoder	340M
CLOP text projector	MLP [1024→1024→1024→512]	1.58M
CLOP cell projector	MLP [512→512→512→512]	1.44M
DiT1D	8× AdaLN-Zero blocks, 8 heads	22.10M
scGPT encoder (frozen)	Transformer	∼51M
scGPT decoder (frozen)	Gene token prediction head	included

Appendix B. CFG Scale Sweep

Table A2 reports the full CFG sweep across 8 guidance scales with 10-step Euler integration and 50 evaluation groups.

Table A2. CFG scale sweep (10-step Euler, 50 eval groups).

CFG	KNN-1	Steering	DivR	KNN/Random
0.5	0.147	0.788	1.13	15×
1.0	0.348	0.796	0.76	35×
1.5	0.396	0.798	0.60	40×
2.0	0.407	0.806	0.53	41×
2.5	0.408	0.804	0.49	41×
3.0	0.410	0.802	0.48	41×
4.0	0.402	0.804	0.50	40×
5.0	0.380	0.798	0.55	38×

Appendix C. Training Hyperparameters

Table A3 lists the complete training hyperparameters for both CLOP alignment and DiT flow-matching stages, corresponding to `configs/clop_v9.3.yaml` and `configs/dit.yaml`.

Appendix D. Supplementary Tables

Table A4. All 80 GEO datasets used in CLOP-DiT training and validation. Datasets were preprocessed to 2,000 highly variable genes per dataset. Split: Train or held-out Validation (study-level, seed 42).

#	Accession	Description	n_{cells}	Split
1	GSE103867	scRNA-seq (H. sapiens)	2,787	Val
2	GSE110686	scRNA-seq (H. sapiens)	2,992	Train
3	GSE115571	Mouse cells, LPS stimulation	442	Train
4	GSE117988	PBMCs from patients	2,638	Train
5	GSE117988	Merkel cell carcinoma tumor	2,990	Train
6	GSE120505	Aged blood cells	3,000	Train
7	GSE123813	Human basal cell carcinoma	3,000	Train
8	GSE123813	Human cutaneous squamous cell carcinoma	3,000	Train
9	GSE123902	Human lung adenocarcinoma	2,753	Train

Continued on next page

#	Accession	Description	n_{cells}	Split
10	GSE124310	Human multiple myeloma bone marrow	2,833	Train
11	GSE130148	Human fetal lung development	3,000	Train
12	GSE132509	Pediatric bone marrow mononuclear cells	2,984	Train
13	GSE138709	Human intrahepatic cholangiocarcinoma	2,891	Train
14	GSE142653	Human pituitary gland development	2,885	Train
15	GSE143423	Human leptomeningeal brain metastasis	3,000	Train
16	GSE143423	Triple-neg. breast cancer brain met.	2,332	Train
17	GSE145929	Adult mouse prostate	2,830	Train
18	GSE145929	Adult mouse urethra	2,745	Train
19	GSE148218	Human bone marrow, ALL	2,807	Train
20	GSE149655	Human early-stage lung adenocarcinoma	2,087	Train
21	GSE155109	Endothelial cells, breast cancer	3,000	Train
22	GSE155109	Stromal cells, breast cancer	3,000	Train
23	GSE163558	Human stomach cancer	2,467	Train
24	GSE165784	Human retina development	2,622	Train
25	GSE165844	Mouse LSK hematopoietic progenitors	2,881	Train
26	GSE167597	Mouse spinal cord development	3,000	Val
27	GSE168181	Human breast cancer tissue	2,863	Train
28	GSE175975	scRNA-seq (H. sapiens)	2,728	Train
29	GSE183904	Human gastric cancer	3,000	Val
30	GSE189070	Mouse astrocytes, spinal cord injury	2,960	Train
31	GSE189357	Human lung adenocarcinoma	2,886	Train
32	GSE192857	hESC differentiation time-series	2,696	Train
33	GSE212502	Human forebrain organoids	3,000	Train
34	GSE213740	Human ascending aortic wall	2,704	Val
35	GSE220913	Human HL-60 cell lines, CFTR	2,972	Train
36	GSE222002	T cells from human cancer	2,962	Train
37	GSE222369	NK cells, human lymphoma	2,778	Train
38	GSE225600	Human breast cancer tumor	2,333	Train
39	GSE225857	Colorectal cancer liver metastases	2,986	Train
40	GSE226131	Aged mouse hematopoietic stem cells	2,608	Train
41	GSE226762	scRNA-seq (H. sapiens)	830	Val
42	GSE227644	scRNA-seq (H. sapiens)	3,000	Train
43	GSE227719	Mouse KrasG12D/P53 lung organoids	2,990	Train
44	GSE228499	Human breast cancer	2,448	Train
45	GSE235787	B cells, ALL	2,798	Train
46	GSE236565	Mouse splenic CD4+ T cells	2,556	Train
47	GSE248214	scRNA-seq (H. sapiens)	2,998	Train
48	GSE253355	Human bone marrow niche	2,835	Train

Continued on next page

#	Accession	Description	n_{cells}	Split
49	GSE262288	Metastatic breast cancer	2,321	Train
50	GSE272187	scRNA-seq (H. sapiens)	2,639	Train
51	GSE275119	Mouse dental tissue development	2,963	Train
52	GSE277740	Mouse thalamus, SBS vs DNBSEQ	2,644	Val
53	GSE279781	scRNA-seq (H. sapiens)	2,654	Train
54	GSE280847	CD45+ leukocytes, mouse MC38 tumor	2,864	Train
55	GSE283205	Hepatoblastoma tumor tissue	2,994	Train
56	GSE283397	Human CD8+ T cells, chronic HBV	2,856	Train
57	GSE288211	Mouse pancreatic islets, Zzef1 KO	2,724	Train
58	GSE289611	Human pulmonary pleomorphic carcinoma	2,983	Train
59	GSE291166	Mouse mesenteric lymph node	2,901	Train
60	GSE299623	scRNA-seq (H. sapiens)	2,991	Train
61	GSE300862	Mouse Kras/Trp53 lung adenocarcinoma	2,716	Val
62	GSE302433	Mouse B16F10 melanoma, ZDHHC13	2,953	Val
63	GSE303309	Brain CD45+ immune cells, Alzheimer's	2,999	Train
64	GSE306567	Mouse ESCs and neural progenitors	2,454	Train
65	GSE306676	Mouse spinal cord, SOD1-G93A ALS	2,959	Train
66	GSE307261	Human B and T cells, melanoma	2,921	Train
67	GSE307774	Mouse colorectal tumors, Kras/Braf	2,505	Train
68	GSE308428	Mouse hippocampal microglia, ASD	2,991	Train
69	GSE309368	Human myeloid cells, healthy donors	1,361	Train
70	GSE315628	scRNA-seq (M. musculus)	2,895	Train
71	GSE318353	scRNA-seq (H. sapiens)	2,964	Train
72	GSE98638	Human hepatocellular carcinoma T cells	3,000	Train
73	GSE120446	Bone marrow, hematopoietic development	2,890	Train
74	GSE104323	Dentate gyrus, brain development	3,000	Train
75	GSE75748	Endoderm differentiation	2,531	Train
76	GSE144024	Human embryonic stem cells	3,000	Train
77	GSE72857	Hematopoiesis	3,000	Train
78	GSE226824	IFN-treated hematopoietic progenitors	2,739	Train
79	GSE141259	Lung development	3,000	Train
80	GSE139369	Human hematopoiesis (Setty et al.)	2,995	Train
	Total		**220,304**	

Table A3. Training hyperparameters.

Hyperparameter	**CLOP**	**DiT**
Optimizer	AdamW	AdamW (β_1=0.9, β_2=0.999)
Learning rate	5×10^{-4}	2×10^{-4}
Weight decay	0.06	0.01
Batch size	1024	1024
Epochs	60	200
Warmup epochs	5	5
LR schedule	Cosine with warmup	Cosine with warmup
Gradient clipping	1.0	1.0
Mixed precision	FP16 (AMP)	FP16 (AMP)
Dropout	0.2	Proj: 0.1, Attn: 0.0
EMA decay	—	0.9999
Temperature	14.0 (fixed)	—
Loss	PrototypeSigLIP	Flow matching MSE
Cohesion weight	0.15	—
Label smoothing	0.1	—
CFG drop prob	—	0.15
Time sampling	—	Logit-normal (μ=0, σ=1)

Table A5. The 8 held-out validation datasets (study-level split, seed 42, val_split = 0.1). No cells from these studies were seen during CLOP or DiT training.

#	Accession	Description	n_{cells}
1	GSE103867	scRNA-seq (H. sapiens)	2,787
2	GSE167597	Mouse spinal cord development	3,000
3	GSE183904	Human gastric cancer	3,000
4	GSE213740	Human ascending aortic wall	2,704
5	GSE226762	scRNA-seq (H. sapiens)	830
6	GSE277740	Mouse thalamus, SBS vs DNBSEQ	2,644
7	GSE300862	Mouse Kras/Trp53 lung adenocarcinoma	2,716
8	GSE302433	Mouse B16F10 melanoma, ZD-HHC13	2,953
	Total		**20,634**

Table A6. Full CFG scale sweep. 200 cells generated per group (100 groups). KNN: type-identity accuracy; Steer.: alignment; DivR: diversity ratio (1.0 = ideal); LinAcc: linear classifier accuracy; FD: Fréchet distance.

CFG	Solver	Steps	KNN-1	KNN-5	Steer.	DivR	LinAcc	FD
Primary configurations (Table 1)								
3.0	Euler	10	0.367	0.554	0.802	0.473	0.508	3.46
2.0	Euler	10	0.369	0.553	0.810	0.513	0.511	3.17
1.0	Euler	10	0.313	0.483	0.810	0.745	0.390	2.84
0.0	Euler	10	0.010	0.051	0.475	1.833	0.010	0.58
3.0	Midpoint	10	0.340	0.528	0.792	0.628	0.477	3.38
2.0	Midpoint	10	0.340	0.525	0.800	0.686	0.475	3.13
1.0	Midpoint	10	0.288	0.461	0.807	0.929	0.357	2.64
3.0	Euler	50	0.348	0.545	0.795	0.622	0.489	3.33
1.0	Euler	50	0.293	0.466	0.807	0.919	0.365	2.65
—	Gaussian	—	0.011	0.048	0.466	2.277	0.009	0.03
Fine-grained CFG sweep (Euler solver)								
0.5	Euler	10	0.147	0.284	0.788	1.129	—	2.33
1.0	Euler	10	0.348	0.515	0.796	0.757	—	2.94
1.5	Euler	10	0.396	0.553	0.798	0.600	—	3.36
2.0	Euler	10	0.407	0.580	0.806	0.526	—	3.54
2.5	Euler	10	0.408	0.590	0.804	0.493	—	3.70
3.0	Euler	10	0.410	0.592	0.802	0.484	—	3.80
4.0	Euler	10	0.402	0.586	0.804	0.502	—	4.01
5.0	Euler	10	0.380	0.576	0.798	0.548	—	4.34
0.5	Euler	20	0.137	0.269	0.788	1.230	—	2.07
1.0	Euler	20	0.334	0.504	0.794	0.842	—	2.83
1.5	Euler	20	0.380	0.545	0.798	0.679	—	3.19
2.0	Euler	20	0.391	0.569	0.798	0.600	—	3.43
2.5	Euler	20	0.394	0.578	0.800	0.559	—	3.66
3.0	Euler	20	0.398	0.582	0.800	0.540	—	3.77
4.0	Euler	20	0.388	0.578	0.802	0.535	—	3.97
5.0	Euler	20	0.379	0.567	0.792	0.551	—	4.16
0.5	Euler	50	0.132	0.259	0.792	1.318	—	1.95
1.0	Euler	50	0.324	0.495	0.794	0.933	—	2.72
1.5	Euler	50	0.370	0.538	0.796	0.776	—	3.14
2.0	Euler	50	0.384	0.563	0.796	0.699	—	3.31
2.5	Euler	50	0.388	0.576	0.796	0.658	—	3.48
3.0	Euler	50	0.392	0.581	0.792	0.637	—	3.64
4.0	Euler	50	0.379	0.579	0.790	0.622	—	3.86
5.0	Euler	50	0.378	0.570	0.790	0.626	—	4.10

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